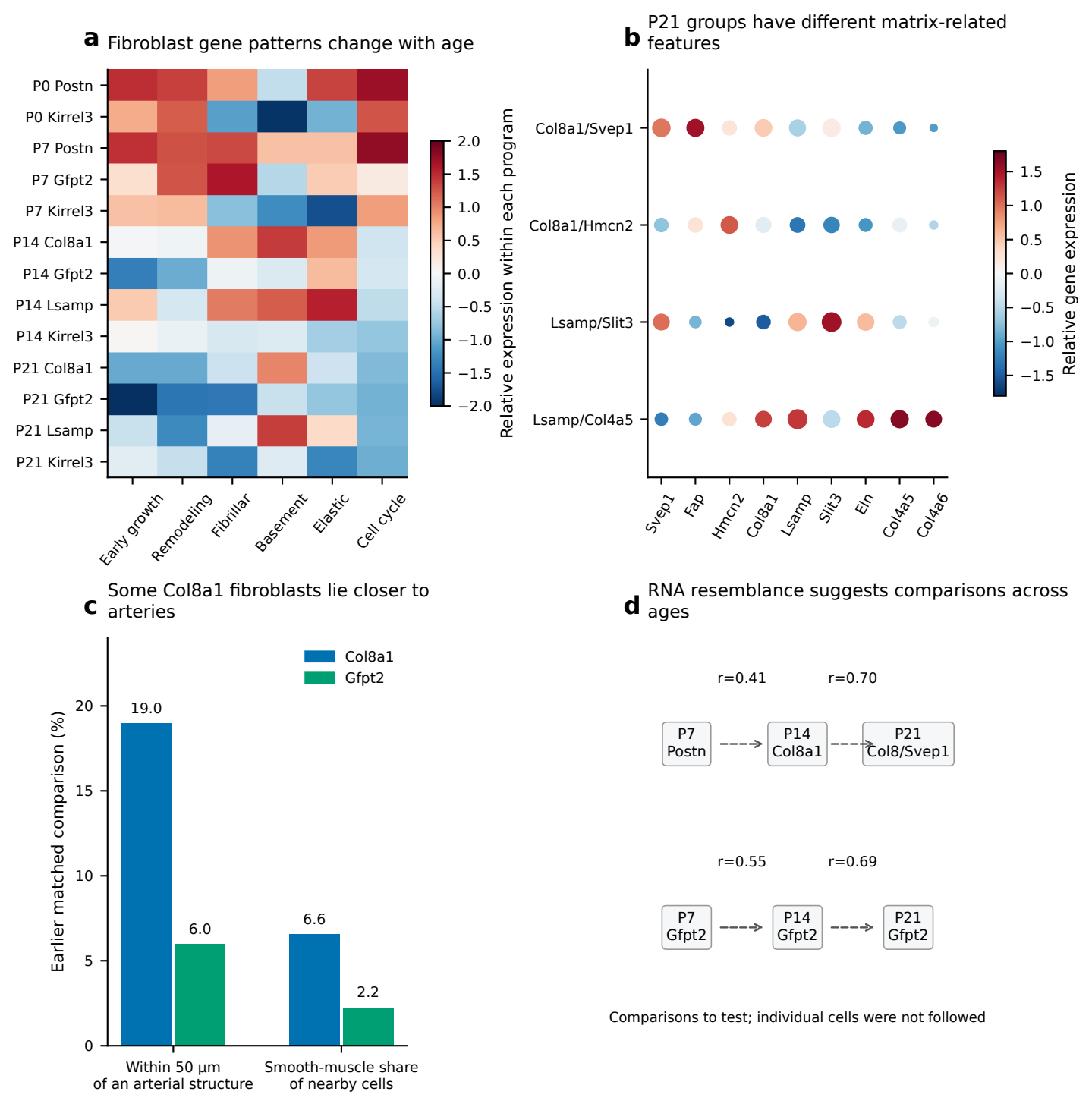
a
Fibroblast gene patterns change with age
P0 Postn
P0 Kirrel3
P7 Postn
P7 Gfpt2
P7 Kirrel3
P14 Col8a1
P14 Gfpt2
P14 Lsamp
P14 Kirrel3
P21 Col8a1
P21 Gfpt2
P21 Lsamp
P21 Kirrel3
Early growth
Remodeling
Fibrillar
Basement
Elastic
Cell cycle
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Relative expression within each program
b
P21 groups have different matrix-related features
Col8a1/Svep1
Col8a1/Hmcn2
Lsamp/Slit3
Lsamp/Col4a5
Svep1
Fap
Hmcn2
Col8a1
Lsamp
Slit3
Eln
Col4a5
Col4a6
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Relative gene expression
c
Some Col8a1 fibroblasts lie closer to arteries
Col8a1
Gfpt2
Earlier matched comparison (%)
0
5
10
15
20
19.0
6.0
6.6
2.2
Within 50 µm of an arterial structure
Smooth-muscle share of nearby cells
d
RNA resemblance suggests comparisons across ages
r=0.41
r=0.70
P7 Postn
P14 Col8a1
P21 Col8/Svep1
r=0.55
r=0.69
P7 Gfpt2
P14 Gfpt2
P21 Gfpt2
Comparisons to test; individual cells were not followed